

Supplementary materials: Estimating Cuba's 2021–2026 population decline

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These materials accompany the main preprint. Models A–D are scenarios; Model D is an illustrative central path, not an identified estimator. Model E is provisional and is not preferred for public claims [Pérez-Riverol, 2026b,a].

1 Life expectancy series

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20 (Figure 1). An independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023, Brønnum-Hansen and Albizu-Campos Espiñeira, 2023]. Life expectancy re-expresses the same mortality shock rather than providing independent corroboration of Model D; its magnitude is large even relative to COVID-era losses documented internationally [Aburto et al., 2022, Msemburi et al., 2023].

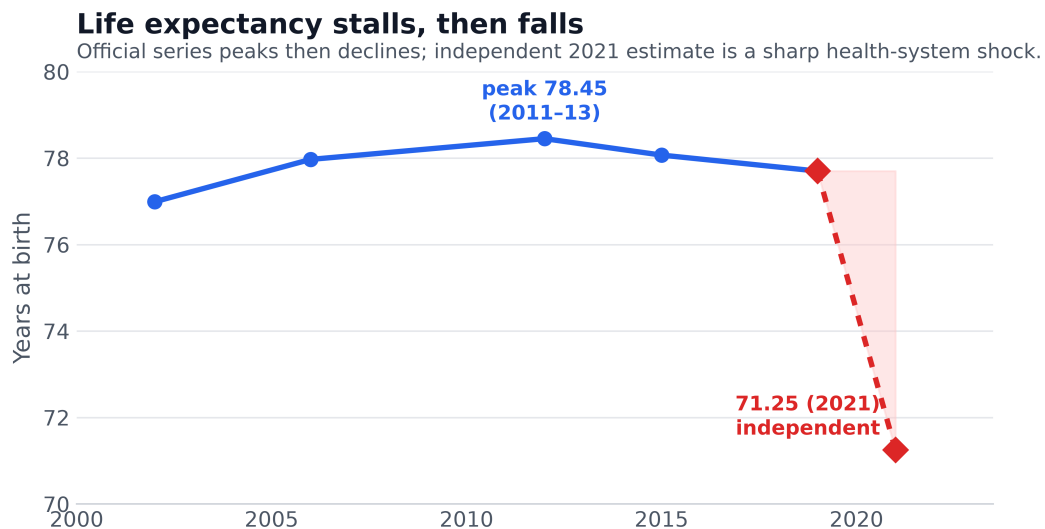


Figure 1: Life expectancy at birth: official triennial series and independent 2021 estimate.

2 Provincial intensity of exit

Emptying is national but unequal (Figure 2). ONEI *Indicadores 2025* reports Havana external net emigration near **−33 per 1,000** in 2025; western and central provinces empty faster than the east [ONEI, 2025]. If national migration accounting is biased, provincial *ranks* are more trustworthy than precise levels.

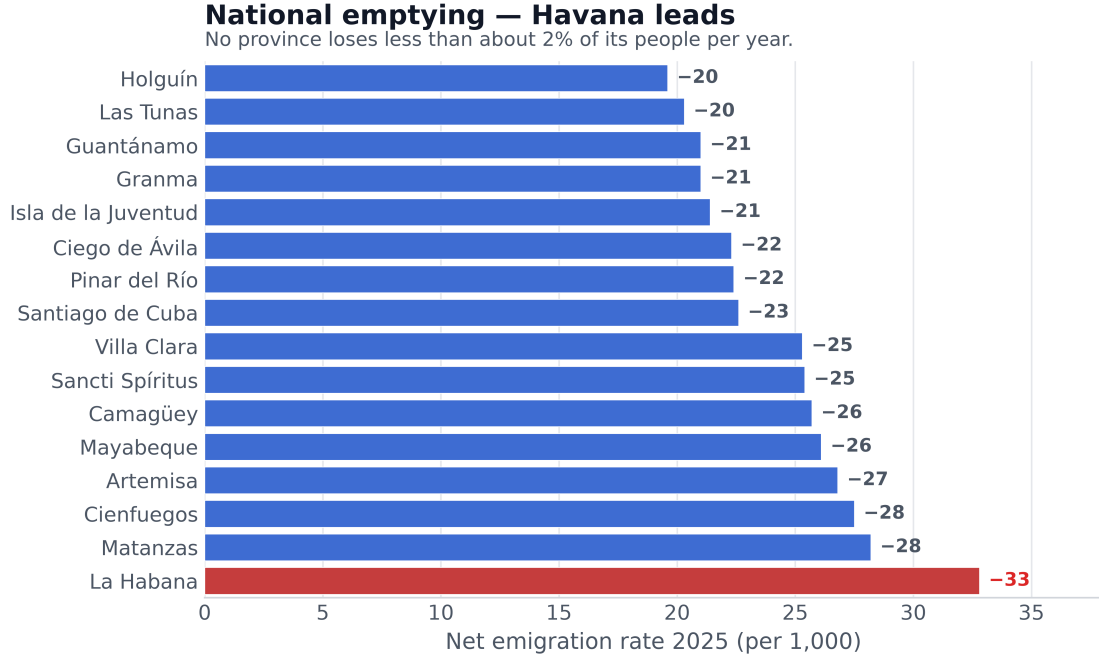


Figure 2: Net emigration rate by province, 2025 (ONEI-reported).

3 Model E: vital reconstruction (provisional)

Model E reconstructs deaths and births from a yearly health-system deterioration score S_t (ageing, health-system decadence, documented underreporting), then closes the identity with a Model D-like migration bridge. The spirit follows excess-mortality reconstruction under incomplete registration [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018], but uses a Cuban-specific score rather than a pandemic-only counterfactual.

Adversarial gate (why E is not promoted).

- Ageing enters both the expected-death schedule and the S_t bucket (double-count risk).
- “Ageing schedule” is currently a two-factor CDR proxy, not $\sum_a m_a P_a$.
- Model E’s U_0 prior is much smaller than Model D’s; the higher end-2025 stock (~ 8.96 M vs 8.59 M) is therefore partly a *different baseline prior*, not independent vital evidence.
- P^* remains migration-identified.

Median end-2025 population about **8.96 million** (loss 2.13 million, 19.2% on the corrected base; migration share $\sim 88\%$). Figure 3 shows S_t and reconstructed versus registered vitals; Figure 4 places E beside A–D. Preferred public claims remain Model D until this gate is cleared [Pérez-Riverol, 2026b].

Table 1: Models A–E medians (end-2025).

Model	Main assumption	Loss (M)	%	Pop. (M)
A · conservative	ONEI-anchored	1.80	16.4	9.16
B · crisis-adjusted	under-reg. + independent migration	2.19	20.4	8.57
C · upper stress	analog upper bound	2.41	22.6	8.26
D · sentinel (illustrative)	IMR check + migration priors near B	2.21	20.5	8.59
E · vital reconstr. (prov.)	HSDS $\rightarrow B^*, D^*$	2.13	19.2	8.96

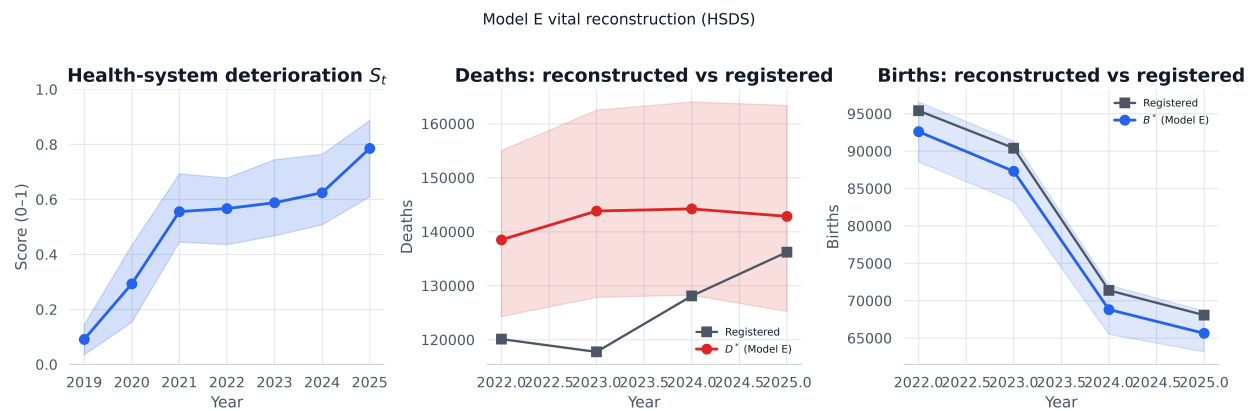


Figure 3: Health-system deterioration score S_t and Model E reconstructed deaths/births versus registered series (2022–2025).

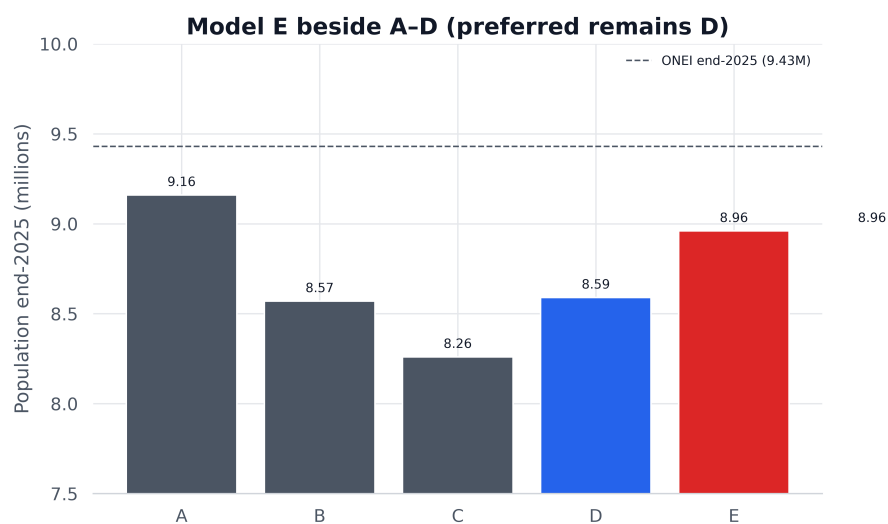


Figure 4: End-2025 population medians for Models A–E. Illustrative central scenario remains D (blue); E (red) is provisional.

4 Data pointers

Scripts and JSON artifacts for HSDS / Model E live under `demographics/scripts/hsds_*.py` and `demographics/artifacts/` in the companion repository [Pérez-Riverol, 2026b]. Adversarial notes: `artifacts/model_e_adversarial_notes.md`.

References

- José Manuel Aburto, Jonas Schöley, Ilya Kashnitsky, Luyi Zhang, Charles Rahal, Trifon I. Missov, Melinda C. Mills, Jennifer B. Dowd, and Ridhi Kashyap. Quantifying impacts of the COVID-19 pandemic through life-expectancy losses: A population-level study of 29 countries. *International Journal of Epidemiology*, 51(1):63–74, 2022. doi: 10.1093/ije/dyab207.
- Juan Carlos Albizu-Campos Espiñeira. Independent estimates of cuba’s resident population using electoral rolls. Technical report, Working paper / report, 2023.
- Henrik Brønnum-Hansen and Juan Carlos Albizu-Campos Espiñeira. Life expectancy in cuba. *Journal of Population Research*, 2023. doi: 10.1007/s12546-023-09296-w.
- Ariel Karlinsky and Dmitry Kobak. Tracking excess mortality across countries during the COVID-19 pandemic with the World Mortality Dataset. *eLife*, 10:e69336, 2021. doi: 10.7554/eLife.69336.
- Nishant Kishore, Domingo Marqués, Ayesha Mahmud, Mathew V. Kiang, Irmay Rodriguez, Arlan Fuller, Peggy Ebner, Cecilia Sorensen, Fabio Racy, Jay Lemery, Leslie Maas, Jennifer Leaning, Rafael A. Irizarry, Satchit Balsari, and Caroline O. Buckee. Mortality in Puerto Rico after Hurricane Maria. *New England Journal of Medicine*, 379:162–170, 2018. doi: 10.1056/NEJMs1803972.
- William Msemburi, Ariel Karlinsky, Victoria Knutson, Serge Aleshin-Guendel, Somnath Chatterji, and Jon Wakefield. The WHO estimates of excess mortality associated with the COVID-19 pandemic. *Nature*, 613:130–137, 2023. doi: 10.1038/s41586-022-05522-2.
- ONEI. Indicadores demográficos 2025 and anuario demográfico tables. <https://www.onei.gob.cu/poblacion-0>, 2025.
- Yasset Pérez-Riverol. Model e vital reconstruction (HSDS) — supplementary materials. File `demographics/manuscript/en/cuba-depopulation-2021-2026-supplement.pdf` in Pérez-Riverol [2026b], 2026a.
- Yasset Pérez-Riverol. Cubascience demographics companion repository. <https://github.com/ypriverol/cubascience>, 2026b.